

RaceXL – Excel Score Sheet Generator

Technical Documentation

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A version of this technical documentation in the Hebrew language can be found at:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Documentation-heb.pdf>

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1. Introduction

The tradition of the “Race to the Technion” is an annual competitive scavenger hunt and puzzle-solving event held at the Technion – Israel Institute of Technology by the Odyssey Youth Program. It requires structured score tracking across multiple activities, with different scoring methods for each activity.

Traditionally, the score sheets have been constructed manually after the race concluded, and the final scores were calculated by hand. This is time-consuming and error-prone, and always results in a delay before the winners are announced, often causing the event to end after the allotted time slot.

“*RaceXL – Excel Score Sheet Generator*” is a utility application designed to automate and streamline the creation of structured Excel score sheets for the “Race to the Technion” event.

This application provides a programmatic solution that generates Excel workbooks containing standardized scoring templates tailored to event-specific activities and scoring rules, based on a JSON configuration file.

The app addresses the challenges stated above by:

- Automatically generating an Excel scores file.
- Supporting customizable configurations.
- Providing pre-built formulas for score calculation.
- Enabling “live” score updates via sharing the spreadsheet with the activities stations managers through *Google Sheets*.
- Ensuring fast results announcement.

These workbooks allow fast score calculation and real-time score tracking, improving efficiency, reducing manual errors, and providing a scalable solution for future event organizers. Additionally, the system is implemented as a command-line tool and is available on Windows and Linux (Ubuntu).

The app is open-source, and the source code, along with all additional information, is posted on GitHub at the following repository:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker>

2. Key Features

- **Scalability:** The JSON-based configuration system supports a variable number of teams and checkpoints to accommodate fluctuating event sizes.
 - **Automatic Excel Generation:** Creates structured .xlsx files pre-configured for team and station data.
 - **Multilingual Support:** The activities and the locations can be in both English and Hebrew.
 - **Live Scoring:** The created spreadsheet can be shared with the activities stations managers through Google Sheets, allowing them to update the scores in “real-time”.
 - **Cross-platform:** The app is available on Windows and Linux (Ubuntu).
 - **Logistical Efficiency:** Significantly reduces the "setup time" required by the administrative team before the race.
 - **Customization:** Enables a custom output file name.
 - **Safe Execution:** A default configuration ensures a smooth experience.
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3. System Overview

The application generates an Excel file (.xlsx) consisting of:

3.1 Scores Sheet

- Centralized summary of all teams.
- Automatic calculation of total scores.
- Integration of race completion time.

3.2 Activity Sheets

- Each activity can have a separate score sheet, in which the station's manager can input the team's results. This is what enables "live" scoring updates.
 - Input methods:
 - Checkboxes
 - Numeric values
 - The separate score sheet is automatically linked to the main score sheet via Excel formulas, so updating it will update the main score sheet too.
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4. Execution

4.1 Windows

Two execution methods:

1. Double-click executable file (recommended).
2. Run via terminal (cmd / PowerShell) – same as Linux.

4.2 Linux (Ubuntu)

Run via terminal, as specified later in sections (5) and (6).

5. Command-Line Interface

The app is a command-line tool and uses Unix-style commands.

- Using the double-click executable on Windows – no need to type ./RaceXL at the beginning.
- Using the terminal on Windows – make sure to reverse the / (replace with \) and to add .exe if needed.
- When naming the output file, the file extension .xlsx is added automatically if omitted.
- The JSON configuration file can be provided without the .json file extension, it is added automatically if omitted.
- In both the output filename and configuration file, relative paths to the execution directory are supported as well.

5.1 General Syntax

```
./RaceXL [help] [name] [configuration]
```

5.2 Flags and Options

Help Flag – display usage instructions

```
--help  
-h
```

- This must be the first flag if used.

Output Naming – set custom output filename

```
--name <filename>  
-n <filename>
```

Default Configuration

```
--default  
-d
```

- Use built-in configuration, skipping additional user prompt.
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6. Usage Scenarios

Under each scenario, all the different ways to achieve it are listed.

6.1 Help Option – Display the Help Menu

- » `./RaceXL --help`
- » `./RaceXL -h`

6.2 Default Execution

- » `./RaceXL`

Use the default filename and prompt whether to use the default configuration.

6.3 Default Configuration (No Prompt)

- » `./RaceXL --default`
- » `./RaceXL -d`

Use the default filename and configuration.

6.4 Custom Configuration

- » `./RaceXL config-file`

Use the default filename and the provided configuration.

6.5 Custom Output File Name

- » `./RaceXL --name output-file`
- » `./RaceXL -n output-file`

Use the provided filename and prompt whether to use the default configuration.

6.6 Custom Output File Name & Default Configuration (No Prompt)

- » `./RaceXL --name output-file --default`
- » `./RaceXL -n output-file -d`
- » `./RaceXL --name output-file --default`
- » `./RaceXL -n output-file -d`

Use the provided file name and the default configuration.

6.7 Full Customization

- » `./RaceXL --name output-file config-file`
- » `./RaceXL -n output-file config-file`

Use the provided filename and configuration.

7. Configuration File Format

The system relies on a JSON configuration file with the following structure:

7.1 Root Fields

- **teams** (integer): Number of participating teams.
- **activities** (array): List of activity objects.

7.2 Activity Object Fields

- **name** (string): Activity name, can be in Hebrew.
- **location** (string): Activity location, can be in Hebrew.
- **makeWorksheet** (boolean): Create an activity sheet or not (Human Pyramid doesn't need a worksheet). The following fields are relevant only when this field is true.
- **rows** (integer): Number of scoring rows.
- **numberTable** (boolean): Activity results input type.
 - true → numeric input
 - false → checkbox input
- **pointsPerRow** (integer): Points per "completed row", i.e., per selected checkbox. This field is relevant only when **numberTable** is false.

Note: It is best and safest to specify values for irrelevant fields as well, as shown in the example configuration in section (7.3).

7.3 Example Race & Configuration

The app's GitHub repository contains an example of a race layout and the JSON file that is associated with it at the following two links:

- Example race layout:
 - In English: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Layout-en.pdf>
 - In Hebrew: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Layout-heb.pdf>
 - Associated JSON file: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Config.json>
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8. Technical Dependencies & Credits

The project integrates the **OpenXLSEX** library, which enables:

- Creation of .xlsx files
- Reading and writing spreadsheet data
- Formula integration

OpenXLSEX is an open-source C++ library for manipulating Microsoft Excel® files in the .xlsx format, licensed under the **BSD 3-Clause License**.

Original code can be found at: <https://github.com/troldal/OpenXLSEX>.

9. License

Copyright © 2026 Amit Zlatkin.

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The usage disclaimer and additional license information are in the app's GitHub repository.

10. Conclusion

The app *RaceXL* provides a fast, robust, and scalable solution for managing the scoring of *The Race to the Technion*. Its flexible configuration system and automation capabilities significantly reduce manual workload while improving accuracy, efficiency, and consistency.